Ashoka University

CS-3410 / MAT-3211-1 · Introduction to Machine Learning · Monsoon 2026

Midterm Project Guide

Three research tracks in classical machine learning

plus an optional advanced SPINN benchmark challenge

Biology · Behavioural Economics · Weather and Climate

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1 Overview and rules

The midterm project asks you to pose a question that data can answer, to answer it with classical machine learning, and to show that your answer holds up. You choose *one* of three research tracks. Each track has its own dataset, built by the teaching team and released on Kaggle. You work on and submit your own project. You also belong to a team of three, used only for a separate team leaderboard. An optional advanced challenge based on a scientific-computing benchmark runs alongside the core project (Section 10). It does not replace the core project.

Table 1: The confirmed structure of the midterm project.

Component	Rule
Tracks	Three tracks: **A** Biology (plant stress), **B** Behavioural Economics (risky choice), **C** Weather and Climate (extreme rainfall). Each student chooses *exactly one*. You do not need to attempt the other tracks.
Submission	**Individual.** Every student submits their own code, experiments, report and prediction file, and sits their own viva.
Teams	Teams of three students. **All three members must choose the same track.** A team exists only for the team leaderboard. It does not turn the project into group work. One member registers the team, with all three Kaggle usernames, on the sign-up form `https://forms.gle/1Rovcnb9DzRga1Dj8` (Ashoka Google account) by Tuesday 20 October 2026, 23:59 IST.
Leaderboards	Your individual leaderboard rank earns *no* regular project marks. The **top three teams** on the team leaderboard receive bonus points, shown separately from the 100-point project score. Bonus amounts: 1st team +5, 2nd team +3, 3rd team +2 marks for each member.
Methods	**Classical machine learning only.** Neural networks, deep learning, pretrained models and foundation models are not permitted in the three core tracks (Section 6.1).
Optional challenge	A separate SPINN-inspired Poisson benchmark challenge in which deep learning, PINNs, numerical solvers and hybrids *are* permitted (Section 10).

> **Key rule.** Team discussion is encouraged. Everything you submit, however, must be your own work, and you must be able to explain it without help. You will be asked to justify your choices and interpret your results in an individual viva.

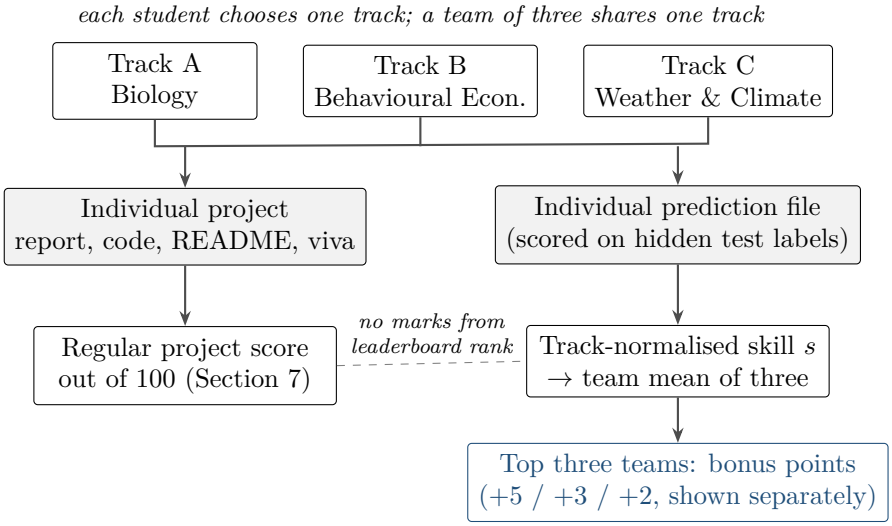


Figure 1: How the parts fit together. Your 100-point project score comes from your individual work. Leaderboard outcomes affect only the separate team bonus.

1.1 Key dates

The project runs for fourteen days; Section 8 gives the day-by-day plan.

Event	Date
Project and datasets released (Day 1)	Saturday 17 October 2026
Team and Kaggle-username registration closes	Tuesday 20 October 2026
Final individual submission (Day 14)	Friday 30 October 2026, 23:59 IST
Individual demonstrations and vivas	Monday 2 – Friday 6 November 2026
Leaderboards close (private leaderboard revealed)	Friday 30 October 2026, 23:59 IST
Optional SPINN challenge deadline	Monday 2 November 2026, 23:59 IST

2 The three datasets

2.1 What the data are, and what they are not

All three datasets are **simulated**. The teaching team wrote a documented simulator for each track, grounded in published scientific findings (plant stress physiology, prospect theory, and the seasonal and synoptic structure of heavy rainfall). A simulator gives you a problem with a known ground truth, honest held-out test labels, and deliberately planted difficulties: shifts between training and test conditions, leakage traps, informative missingness and rare events. Real data rarely offer all of these in a dataset of manageable size.

> **Interpretation boundary.** No row in any of the three datasets is a real plant, a real person or a real weather observation. Your conclusions are about the simulated system and about the methods you used. Do not present them as agronomic, behavioural or climate findings. Say *simulated* whenever you describe the data in your report.

Each Kaggle dataset page contains the same files. The page also states, in prose, how the data were simulated. The exact simulator coefficients are withheld so that the test set stays meaningful.

File	Contents
`train.csv`	Features and the target.
`test.csv`	Features only. Test labels are private and never released.
`sample_submission.csv`	The exact prediction-file format for the leaderboard.
`README.md`	Data dictionary with units, the simulation summary and the known pitfalls.

2.2 Design of the training and test split

In every track, the test set differs from the training set *on purpose*, in the way that matters for that track's research question (Table 2). A model that merely memorises the training conditions will be found out. This is why the split you use for your own validation matters as much as the model you choose.

Table 2: Dataset sizes, test-set design and primary metric for each track.

Track	Size (train / test)	How the test set differs from training	Primary metric
A Biology	4,200 / 1,800 plants	Two growing conditions never seen in training, both hotter than any training condition	Macro-averaged F1 (3 classes)
B Behav. Econ.	8,400 / 3,600 decisions	90 participants never seen in training (210 in training, 40 decisions each)	ROC-AUC
C Weather	64,288 / 29,200 station-days	Later years (2021–2024 vs. 2010–2020) and four stations never seen in training	Average precision (PR-AUC)

2.3 Dataset links

Track	Link
A dataset	`https://www.kaggle.com/datasets/samarjeetsinghmalik/iml2026-track-a-simulated-plant-stress`
B dataset	`https://www.kaggle.com/datasets/samarjeetsinghmalik/iml2026-track-b-simulated-risky-choice`
C dataset	`https://www.kaggle.com/datasets/samarjeetsinghmalik/iml2026-track-c-simulated-extreme-rain`
A competition (join link)	`https://www.kaggle.com/t/c460443624204a52b6c12796be312881`
B competition (join link)	`https://www.kaggle.com/t/fc0bc2dabcb04acc9d100158fce95dd5`
C competition (join link)	`https://www.kaggle.com/t/aef8156c4b0d4518a7b5c4679f7ad760`

3 Track A — Biology: predicting plant stress

Working title. *Can physiological measurements reveal plant stress, and does a model trained in some growing conditions still work in others?*

3.1 Background

Plants under water deficit close their stomata to conserve water. Transpiration falls, the canopy warms, and leaf water content and photosynthesis decline [1]. Heat stress damages membranes and the photosynthetic apparatus. A well-watered plant may keep its stomata open and cool itself by transpiration, so heat and drought can leave different physiological fingerprints [2]. Breeders and agronomists would like to read stress from cheap, fast measurements instead of waiting for yield losses at harvest. The difficulty is that a classifier fitted in one environment may have learned that environment's *particular* route to stress.

3.2 The dataset

The data contain 6,000 simulated plants of five cultivars, measured once each. The simulator combines a drought pathway and a heat pathway into a latent stress score, which is banded into the label `stress_level` ∈ {`none`, `moderate`, `severe`}. Features include the following:

- the environment: air temperature, humidity, vapour-pressure deficit, light, irrigation and soil moisture;
- physiology: stomatal conductance, canopy temperature, relative water content, chlorophyll (SPAD), NDVI, proline and electrolyte leakage;
- growth: leaf area and plant height;
- structure: cultivar, plot and the chlorophyll-meter batch.

The training file covers four growing conditions (two greenhouse and two field regimes). The test file covers two further conditions.

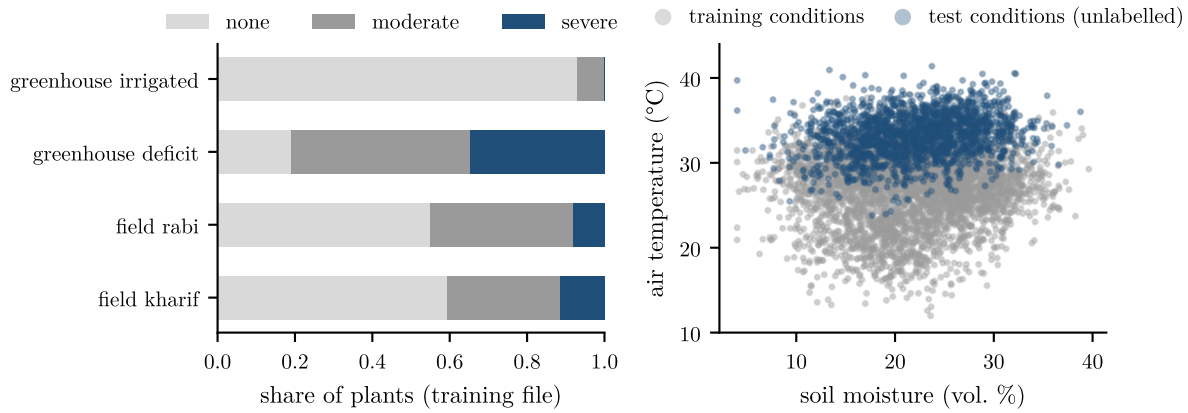


Figure 2: Track A training data. Left: the stress-class mix differs sharply between growing conditions. Right: the unlabelled test conditions (accent) occupy a hotter region of the environment than the training conditions (grey).

3.3 Research questions you can pursue

Choose one focused question. Questions marked ⋆ are more demanding.

1. **Generalisation across conditions.** How much does performance fall when you validate on a growing condition held out from training (leave-one-condition-out), compared with a

random split? Which models degrade least, and why?

2. **Feature relevance.** Which measurements carry the most information? Is a small panel of cheap measurements nearly as good as the full set? Is the selected panel stable across folds?
3. **Measurement problems.** Does the miscalibrated sensor batch bias predictions? Does treating missingness as informative (for example, with missing-value indicators) help?

⋆4. **Two routes to stress.** Do the training conditions contain enough heat-stressed plants to learn the heat pathway? What happens to a model in a hot, well-watered condition?

⋆5. **Ordinal structure.** Does respecting the ordering none $<$ moderate $<$ severe (for example, cumulative binary models) beat treating the classes as unordered?

3.4 Suggested baselines and experiments

- **Baselines** (at least three, Section 6.1): a stratified or majority-class dummy; multinomial logistic regression with L_2 or L_1 regularisation; LDA/QDA or Gaussian Naive Bayes; a linear or kernel SVM.
- **Controlled experiments** (at least two), for example: random versus condition-held-out validation; all features versus a selected panel; with versus without missing-value indicators; class weighting for the minority class.
- **Metric.** Macro-averaged F1 is primary: every class counts equally, whatever its frequency. Also report per-class precision and recall and the confusion matrix.

3.5 Pitfalls built into the data

- `harvest_yield_loss_pct` is measured at harvest, weeks after the stress assessment. It appears only in `train.csv` and predicts the label almost perfectly, which is exactly why using it as a feature is leakage: your model could not be applied to the test set.
- Plants in the same plot share a micro-environment and are not independent. Group-aware validation may be needed.
- Stomatal conductance is missing more often for severely wilted plants. Missingness is not random.
- The class mix in the test conditions differs from the training mix (Figure 2).

4 Track B — Behavioural Economics: predicting risky choices

Working title. *Do framing, incentives and individual differences help predict choices under risk, and do models generalise to new people?*

4.1 Background

Expected-value maximisation predicts that people choose whichever option has the higher average payoff. Decades of experiments show systematic departures from this. People are typically risk-averse for gains, risk-seeking for losses and averse to mixed gambles, and they overweight small probabilities [3, 4]. The *framing* of an identical problem as gains or as losses changes choices [5]. Prediction is also hard for a different reason: people differ from one another, so a model that knows *this* person's history can look far better than one meeting a stranger.

4.2 The dataset

The data contain 12,000 simulated binary decisions: 300 simulated participants make 40 decisions each. Each decision is a choice between a risky prospect (two outcomes with stated probabilities) and a sure amount. The target is `chose_risky`. Problems are in the gain, loss or mixed domain. Gain problems are presented either in a *gain frame* or in a *loss frame*, where the same final-wealth problem is shown as deductions from a larger endowment. Context variables record payment (paid or hypothetical sessions), time pressure, numeracy, age and self-reported risk taking. Choices were generated by a stochastic prospect-theory model with participant-specific parameters.

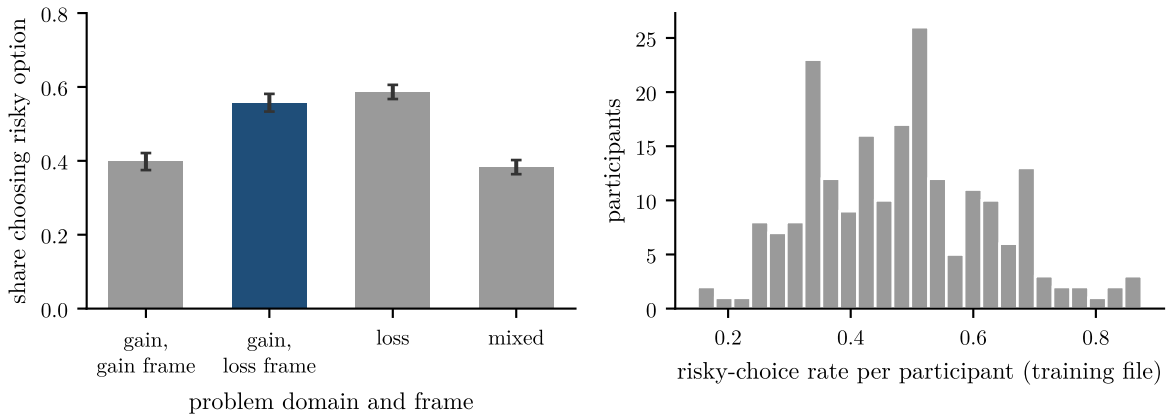


Figure 3: Track B training data. Left: share of risky choices by problem domain and frame (95% intervals). The same gain problems are chosen differently under a loss frame (accent). Right: participants differ widely in how often they take risks.

4.3 Research questions you can pursue

1. **Does framing matter for prediction?** Compare a model using only the nominal attributes (probabilities, outcomes, sure amount) with one that also sees the frame and context. Expected value cannot see the frame. Can your model?
2. **Generalising to new people.** How optimistic is a random row-level split compared with a participant-held-out split (for example, `GroupKFold`)? Which features let a model "recognise" a participant?
3. **A transparent benchmark.** How far does a simple economic rule, such as choosing the option with the higher expected value, get you? Where does a learned model improve on it, and where does it not?

⋆4. **Feature engineering from theory.** Do features motivated by the literature (final wealth, expected-value difference, probability of the larger outcome, domain × frame interactions) help a linear model?
⋆5. **Calibration.** Are your predicted probabilities calibrated [6]? Does calibration differ between paid and hypothetical sessions?

4.4 Suggested baselines and experiments

- **Baselines:** a constant (prior) predictor; the expected-value rule as a transparent benchmark; logistic regression (regularised); LDA or Gaussian Naive Bayes; a linear SVM.
- **Controlled experiments:** nominal versus nominal-plus-frame/context features; random versus participant-held-out validation; interaction features versus none.
- **Metric.** ROC-AUC of the predicted probability is primary. Also report log-loss, accuracy at a stated threshold, and a calibration (reliability) plot.

4.5 Pitfalls built into the data

- `realised_payout_inr` is what the participant was paid *after* choosing. It appears only in `train.csv`, and it almost reveals the choice. Using it is leakage.
- Decisions from the same participant are correlated. A random split puts the same person in both training and validation data.
- `response_time_ms` is recorded with the choice. If you use it, justify why it would be available at prediction time.

> **Ethics and interpretation.** The participants are simulated. Even so, practise the habits you would need with real people. Do not infer personality, welfare or ability from predictions. Do not describe associations as causal effects unless the design supports it.

5 Track C — Weather and Climate: predicting extreme rainfall

Working title. *Can today's weather warn of tomorrow's heavy rain, and does the warning still work in a warmer future and at stations it has never seen?*

5.1 Background

Heavy-rain days are rare but cause most flood damage. Warmer air holds more water vapour, roughly 7% more per degree Celsius (the Clausius–Clapeyron relation). This is one reason heavy precipitation is expected to intensify as the climate warms [7]. For a classifier, rare events change what "good" means. Accuracy becomes almost useless, and the trade-off between catching events and raising false alarms takes centre stage [8]. Because the climate drifts, a model validated on the past may be miscalibrated for the future.

5.2 The dataset

The data contain 20 simulated stations, observed daily from 2010 to 2024. Each row is one station-day of weather:

- temperature, dew point and humidity;
- pressure and its 24-hour change;
- precipitable water and an instability index;
- wind, cloud cover, and today's and recent rain.

The target `heavy_rain_tomorrow` is 1 if the *next* day's rain is at least 64.5 mm, the India Meteorological Department threshold for "heavy rainfall". About 3% of training days are positive. The simulator has a monsoon-like seasonal cycle, slowly varying moisture, travelling disturbances that announce themselves a day ahead (falling pressure, rising humidity and instability), and a warming and moistening trend. The climate is *not* stationary: the balance between the drivers of heavy rain shifts over the years, and some stations had sensors recalibrated during the record.

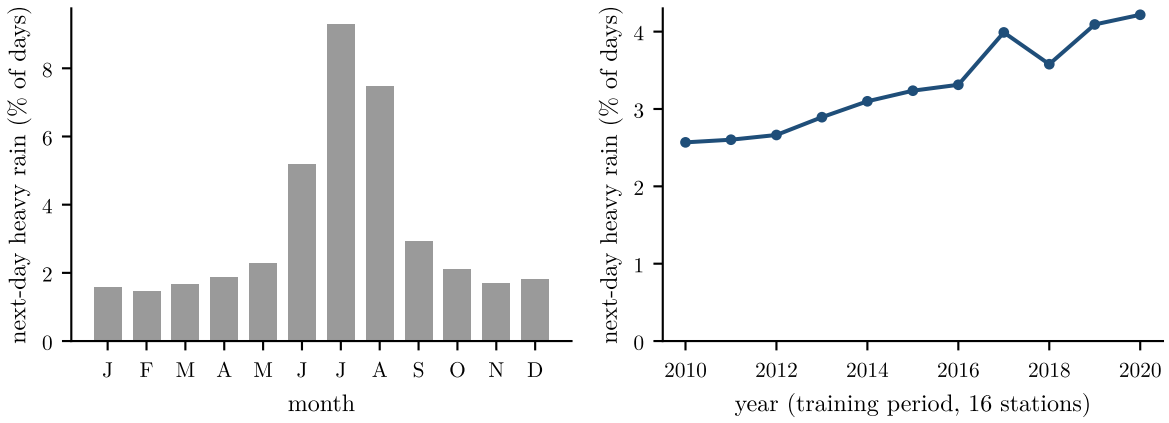


Figure 4: Track C training data. Left: heavy-rain days cluster in the monsoon months. Right: the yearly frequency drifts upwards over the training period. The test period (2021–2024) is later still.

5.3 Research questions you can pursue

1. **Does random splitting flatter the model?** Compare random, blocked-by-time and later-period validation [9]. Report how far the estimates differ, including if they barely differ.
2. **Detection versus false alarms.** Where should the decision threshold sit? Report precision–recall curves and the hit rate and false-alarm ratio at a few operating points.

Argue for one operating point.

3. **Generalisation in space.** Hold out whole stations. Which features transfer between stations, and which encode station identity?

⋆4. **Drift and calibration.** Are your probabilities calibrated on later years? Does adding a trend term, or re-weighting recent years, help or hurt?

⋆5. **Physically motivated features.** Do derived predictors such as dew-point depression or the humidity change from morning to afternoon improve a linear model?

5.4 Suggested baselines and experiments

- **Baselines:** the prevalence (constant) predictor; climatology (the training-period rate for each station and calendar month); regularised logistic regression; LDA or Gaussian Naive Bayes; a linear SVM.
- **Controlled experiments:** random versus chronological validation; class weighting versus none; with versus without station-identity features; threshold selection on validation data.
- **Metric.** Average precision (area under the precision–recall curve) is primary. Also report ROC-AUC, recall and false-alarm ratio at your chosen threshold, and a reliability plot.

5.5 Pitfalls built into the data

- `rain_tomorrow_mm` is tomorrow's rain, the quantity that defines the label. It appears only in `train.csv`. Using it is leakage.
- Consecutive days are strongly dependent. A random split lets the model see the day before and the day after each validation day.
- The relationship between the features and heavy rain drifts over time, and some sensors were recalibrated. A random split therefore *overstates* how well a model will do on later years.
- Every feature must be available by the end of the observation day. Check any feature you construct, such as rolling windows, for look-ahead.
- Instrument outages leave runs of missing values. Do not impute them using future data.

6 Requirements common to all three tracks

6.1 Permitted methods

The core project is about *classical* machine learning. The rule has two parts.

Table 3: Methods policy for the three core tracks.

Category	Policy
Taught in lectures	Linear and polynomial regression; ridge and lasso; discriminant functions and Fisher's LDA; probabilistic generative classifiers (LDA, QDA, Gaussian Naive Bayes); logistic regression; support vector machines. **Your required baselines come from this group, plus one trivial baseline** (dummy, prior or climatology).
Classical, not yet taught	Decision trees, random forests, classical gradient boosting, k-nearest neighbours, and classical feature-selection methods. **Permitted** for additional experiments. You must be able to explain how the method works, and why you chose it, in the viva.
Not permitted	Neural networks of any kind, deep learning, pretrained models, foundation and large language models used as predictors, and AutoML systems that hide the model choice.

Standard libraries are allowed (for example NumPy, pandas, scikit-learn [10] and matplotlib), but you are responsible for every line you submit. The course policy on generative-AI assistance for code and writing is: allowed for coding help and language editing, provided you declare the tool and what you used it for in your README; you must be able to explain every line you submit in the viva, and AI-written analysis must not be presented as your own.

6.2 Minimum experimental standard

1. **One focused, testable research question** or hypothesis, with a sentence on why it matters for your track.
2. **A concise literature review**: three to six sources on the scientific background and the methods, cited properly. The track sections suggest starting points.
3. **At least three baseline models**, chosen as in Table 3, all evaluated with the same protocol.
4. **At least two controlled experiments beyond the baseline comparison.** Each one changes one thing (a feature set, a validation scheme, a weighting, a threshold) and tests a stated expectation.
5. **Justified metrics.** Use the track's primary metric, plus the secondary metrics that your question needs.
6. **Sound evaluation** (Figure 5). Fit everything that learns from data (imputation, scaling, feature selection, the model itself) inside a pipeline, on training folds only. Tune on validation folds that respect the data's structure (conditions, participants, time, stations) [9, 11]. The Kaggle test set is used only for your final prediction file.
7. **Error analysis and limitations.** Which cases fail, and why? What could be leakage, bias or chance [12, 13]?
8. **Reproducibility.** Fixed seeds, a pinned environment, and one documented command that regenerates your key tables and your prediction file.

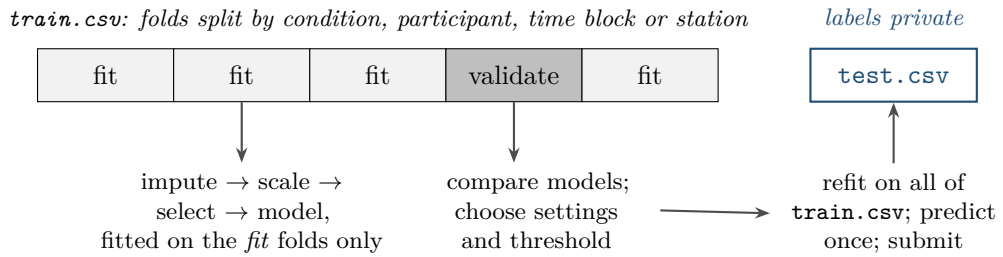


Figure 5: The evaluation protocol. Everything that learns from data is fitted inside the training folds. The test file is touched once, at the end.

6.3 Deliverables

Upload your deliverables to the Google Classroom assignment *IML Midterm Project* by Friday 30 October 2026, 23:59 IST.

1. **Report** (PDF, about 4–6 pages excluding references and appendix), written from the supplied template. Sections: question; background and related work; data and preprocessing; methods and experimental design; results; error analysis and limitations; conclusion.
2. **Code** as a notebook or a small repository, with a `README.md` written from the supplied template. It must contain the environment, the exact commands, the seeds, the expected runtime and a map from each table or figure to the code that produces it.
3. **Prediction file** for your track, in exactly the format of `sample_submission.csv`. Submit it to your track's Kaggle competition, and include the same file in your upload.
4. **A results table** (`experimental_results.csv`, template supplied) listing every run you report.
5. **Individual demonstration and viva**: a short live run of your code, then questions on your choices and results. This applies to every student.

> **Academic integrity.** Cite every dataset, paper, library and piece of adapted code. Discussing ideas within your team is fine. Sharing code or report text is not, because each submission is assessed individually. Your prediction file must be reproducible from the code you submit. A file that cannot be reproduced does not count for the leaderboard and may be referred under the university's academic-integrity procedures.

7 Assessment

7.1 Regular project score (100 points)

The rubric below is the marking scheme for the project. Every component is assessed for each student individually.

Table 4: The 100-point rubric. Leaderboard rank does not enter this score.

Component	Marks	What earns full marks
Research question and literature review	10	A focused, testable question; three to six relevant sources used, not just listed.
Data understanding, preprocessing and leakage prevention	15	Data audited; preprocessing inside the pipeline; train-only and future columns identified and excluded; a validation split that matches the data's structure.
Baselines and experimental design	20	At least three appropriate baselines; at least two controlled experiments, each testing a stated expectation.
Results, error analysis and limitations	25	Correct metrics with uncertainty where feasible; failure cases examined; honest limitations, including the simulated nature of the data.
Code quality and reproducibility	10	One documented command reproduces the key results and the prediction file; seeds and environment pinned.
Report communication and demonstration	10	A clear 4–6 page report, readable figures and tables, and a working live demonstration.
Individual viva and understanding	10	You can explain and defend every modelling choice without help.
Total	**100**	

7.2 Team bonus (separate from the 100 points)

The top three teams on the team leaderboard (Section 9) receive bonus points, recorded separately from the regular score. The amounts are: 1st team +5, 2nd team +3, 3rd team +2 marks for each member. Bonus points never replace any component of the individual assessment.

8 Fourteen-day schedule

The core project runs over fourteen days, from Saturday 17 October 2026 (Day 1) to the final submission on Friday 30 October 2026, 23:59 IST (Day 14). Individual demonstrations and vivas follow in the week of Monday 2 – Friday 6 November 2026. The optional SPINN challenge has its own deadline, Monday 2 November 2026, 23:59 IST (Section 10).

9 Leaderboards

9.1 Individual track leaderboards

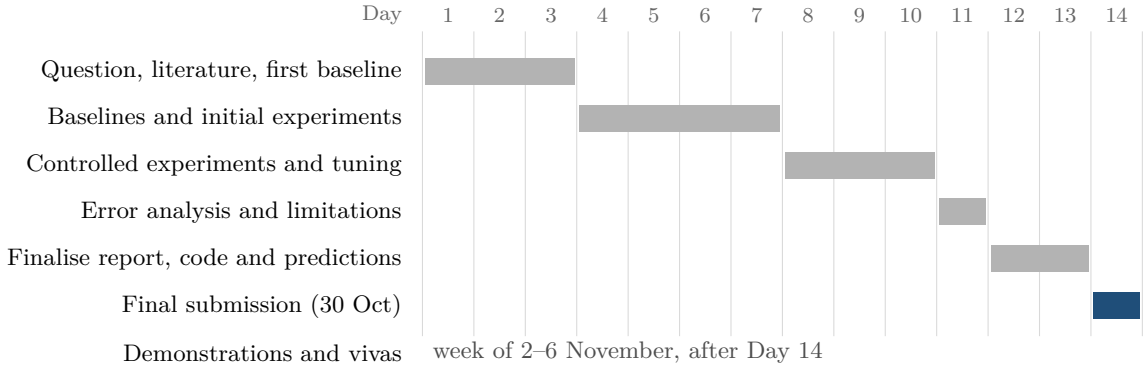
Each track has its own Kaggle competition (links in Section 2). You submit predictions for *your* track, under your own Kaggle account and as a solo participant. Kaggle's own team-merging feature is not used. The competition scores your file against the hidden test labels:

- the **public leaderboard**, shown while the competition runs, uses about 30% of the test rows;
- the **private leaderboard**, revealed when the competition closes, uses the remaining 70% and is the only one that counts.

Register your team and your Kaggle username on the team sign-up Google Form, `https://forms.gle/1Rovcnb9DzRga1Dj8`, by Tuesday 20 October 2026 (23:59 IST), so that your

Table 5: Fourteen-day schedule for the core project (Day 1 = Saturday 17 October 2026).

Days	Milestone
1–3 (17–19 Oct)	Understand the selected dataset, formulate a research question, review literature, and establish a baseline
4–7 (20–23 Oct)	Implement baselines and conduct initial experiments
8–10 (24–26 Oct)	Run controlled experiments and parameter tuning
11 (27 Oct)	Error analysis and limitations
12–13 (28–29 Oct)	Finalise report, code, reproducibility, and prediction submission
14 (30 Oct)	Final submission; individual demonstrations and vivas follow (2–6 Nov)



leaderboard score can be matched to you. Sign in with your Ashoka Google account. One member registers the whole team and can edit the response until the deadline. If the form will not let you register, email the team's details to the teaching fellow by the same deadline. Chasing the public leaderboard is a form of overfitting. Trust your own validation. The daily submission limit is 5 submissions per day, with 1 final submission that you nominate. Your individual rank earns no regular marks.

9.2 Team leaderboard

Raw scores cannot be compared across tracks. A macro-F1 of 0.6 on Track A and an average precision of 0.3 on Track C are not on the same scale, and the tracks differ in how much improvement is achievable. Each student's private-leaderboard score m is therefore converted into a *track-normalised skill*

$$s \;=\; \min\!\left\{\, \max\!\left\{ \frac{m-b}{r-b},\; 0 \right\},\; 2 \right\}, \tag{1}$$

where b is the score of the track's trivial baseline and r is the score of the teaching team's reference model, both computed on the same private test rows. The reference model is a standardised, L_2-regularised logistic regression with default settings, and it is the same on every track. So $s = 0$ means "no better than the trivial baseline" and $s = 1$ means "as good as the reference model" on *every* track. Beating the reference earns more, up to a cap of 2.

- **Team score** = the mean of the three members' s. A member with no valid submission contributes $s = 0$.
- **Ties** (equal team scores) are broken first by the higher median member score, then by the earlier time at which the team's last member made their final valid submission.
- **Top three teams** receive the bonus described in Section 7.

> **Validity.** A leaderboard entry counts only if the submitted code reproduces it. The teaching team may re-run any submission. Test labels are never released, and attempts to obtain them (for example by probing the leaderboard with many crafted submissions) disqualify the entry.

The course competitions are Kaggle *Community* competitions. They do not award Kaggle points, medals or tiers. Upvoting is never part of the assessment. Do not organise or join vote-trading, which Kaggle's Community Guidelines forbid.

The normalisation formula, the baseline and reference definitions, the tie-breaks and the submission limit are fixed before the competitions open. The bonus amounts are 1st team +5, 2nd team +3, 3rd team +2 marks for each member.

10 Optional challenge: beat the SPINN Poisson benchmark

Status. Optional, and separate from the three core tracks. It is *not* a fourth project track. Taking part does not replace or reduce any part of the core project, including its individual viva. Open to third-year, fourth-year and PhD students only. Deadline: Monday 2 November 2026, 23:59 IST. Award: a submission that passes Stages 1 and 2 adds 3 marks to its author's final IML course total for a third-year student, 2 marks for a fourth-year student and 1 mark for a PhD student.

10.1 Origin and objective

The challenge comes from the coding test reported in a GSoC 2026 proposal for the ML4SCI project *Physics-Informed Neural Network Shape Optimization* (SPINN), written by this course's teaching fellow [14]. The proposal solved the Poisson problem below with a physics-informed neural network [15], using a 6×50 tanh network in the configuration of the coordinate-projection framework [16]. It reports four error figures. **Your objective is to beat all four reported values** with a solver that also generalises to problems it has never seen.

10.2 Problem

Find u on the unit square such that

$$\nabla^2 u(x, y) = f(x, y), \qquad 0 < x, y < 1, \qquad u = g \text{ on the boundary.} \tag{2}$$

The goal is a *general* solver: given any forcing f and boundary data g, return u. The public reference case, as in the proposal, has $g = 0$ and the analytical solution and forcing

$$u(x, y) = \sin(\pi x)\sin(\pi y), \qquad f(x, y) = -2\pi^2 \sin(\pi x)\sin(\pi y). \tag{3}$$

10.3 Permitted methods

Unlike the core tracks, **any appropriate method is allowed**:

- PINNs and other neural-network approaches, alternative architectures, and coordinate-based models;
- grid- or Gaussian-based learnable representations, such as PIXEL- or PIG-inspired methods [17, 18];
- classical numerical PDE solvers: finite differences [19], finite elements, finite volumes and spectral methods;
- hybrids of the above.

Whatever you choose, the solver must compute its answer from the inputs it receives (Section 10.8).

10.4 Benchmark values and qualification targets

Table 6 separates what the proposal *reports* (superscript ^{R}) from the rules the organisers *set* for this challenge (superscript ^{O}).

Operational metric definitions^{O}. The proposal does not state its evaluation grid or how its boundary loss was normalised. This challenge therefore fixes its own definitions. Let $\mathcal{G}$ be the uniform $N \times N$ grid on $[0, 1]^2$ including the boundary ($N = 129$ for the public reference case), let $\partial\mathcal{G}$ be its boundary nodes, $\hat{u}$ your prediction and u the exact solution. Then

$$\mathrm{MAE} = \frac{1}{|\mathcal{G}|} \sum_{\mathcal{G}} |\hat{u} - u|, \qquad E_{\max} = \max_{\mathcal{G}} |\hat{u} - u|,$$

$$E_{\mathrm{rel},L_2} = \frac{\|\hat{u} - u\|_{2,\mathcal{G}}}{\|u\|_{2,\mathcal{G}}}, \qquad \mathrm{MSE}_{\mathrm{BC}} = \frac{1}{|\partial\mathcal{G}|} \sum_{\partial\mathcal{G}} (\hat{u} - g)^2,$$

Table 6: Benchmark values from the quantitative-results table on page 6 of the proposal, and the Stage 1 qualification targets.

Metric	Reported result[R]	Strict qualification target[O]
Mean absolute error (MAE)	$2.39 \times 10^{-4} \pm 1.2 \times 10^{-5}$ (five-run mean ± standard deviation)	MAE $< 2.39 \times 10^{-4}$
Maximum absolute error	2.85×10^{-3}	$E_{\max} < 2.85 \times 10^{-3}$
Relative L_2 error	Approximately 10^{-3}	$E_{\mathrm{rel},L_2} < 10^{-3}$
Final boundary-condition loss	Approximately 10^{-6}	$\mathrm{MSE}_{\mathrm{BC}} < 10^{-6}$

[R] Reported in the GSoC 2026 SPINN proposal [14]. [O] Organiser rule for this challenge; not a result from the proposal.

where g is the prescribed boundary value. Comparisons use full floating-point precision; rounded displayed values never decide qualification. Because the proposal's grid is unspecified, beating these values is an operational comparison, not a like-for-like re-run of the original experiment.

> **What the benchmark does and does not measure.** The reported values are strong for a PINN but not for numerical analysis in general. On the 129×129 reference grid, a standard second-order five-point finite-difference solver already beats all four Stage 1 targets by more than an order of magnitude. Its boundary error is exactly zero, because the boundary is imposed directly. Stage 1 is therefore an *entry gate*. The ranking is decided by a much harder hidden suite (Stages 2 and 3), on which a plain second-order solver ranks well below more accurate, more general methods.

10.5 Staged evaluation

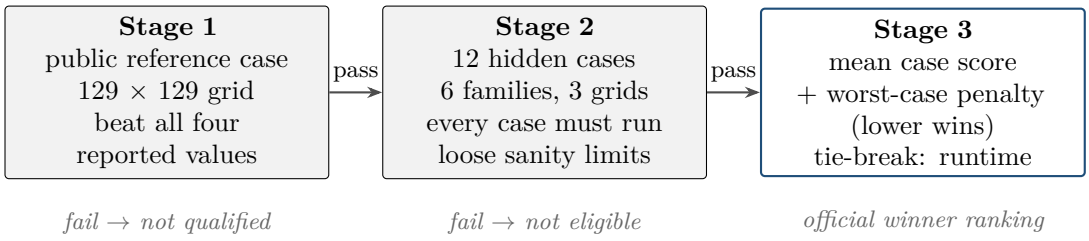


Figure 6: The three evaluation stages. Only submissions that pass Stages 1 and 2 are ranked.

Stage 1: public benchmark qualification. Your solver is run on the public reference case (3) on the 129×129 grid and must be *strictly* below all four targets in Table 6. Trainable or stochastic methods are run five times with seeds $0, \ldots, 4$, passed in as `run_seed`, and qualification uses the five-run mean of each metric. This mirrors the five-run protocol behind the proposal's MAE. The public evaluator runs exactly this check, and also reports six further public practice cases.

Stage 2: hidden generalisation[O]. The teaching team runs every Stage 1 qualifier on a hidden suite of twelve cases that it generates after the deadline from a private seed. Two cases come from each of six families:

- high-frequency sine series (modes up to 12);
- smooth bubble–exponential solutions;
- narrow Gaussian bumps;

- sharp internal layers of width about 0.015–0.03;
- harmonic functions with *non-zero* boundary values g;
- mixtures of the above.

Grids are 65×65, 129×129 and 257×257. The six public practice cases come from the same families, at milder settings. A submission is eligible if every hidden case runs within the time limit and returns a valid array, and its *worst* case satisfies

$$E_{\mathrm{rel},L_2} < 5 \times 10^{-2}, \qquad \mathrm{MSE}_{\mathrm{BC}} < 10^{-4}.$$

These loose limits only screen out solvers that fail on a family. They are organiser rules, not values from the proposal.

Stage 3: winner ranking°. Eligible submissions are ranked by a score that rewards being good on *every* case. For hidden case i,

$$c_i = \sum_k w_k \, \max\left\{ \log_{10} \frac{e_{ik}}{\tau_k}, \, -6 \right\},$$

and the final score (lower is better) is

$$S = \frac{1}{12} \sum_{i=1}^{12} c_i \; + \; 0.25 \, \max_i c_i, \tag{4}$$

where

- e_{ik} is the value of metric k on hidden case i;
- τ_k is the corresponding reported value (2.39×10^{-4}, 2.85×10^{-3}, 10^{-3} and 10^{-6});
- the weights are $w = (0.25, 0.25, 0.35, 0.15)$ for (MAE, $E_{\max}$, E_{rel,L_2}, $\mathrm{MSE}_{\mathrm{BC}}$).

The logarithm puts errors spanning many orders of magnitude on a common scale. The floor of -6 means that being more than 10^6 times below a target earns nothing extra, so no single metric (such as a boundary error of exactly zero) can dominate. The worst-case term penalises a solver that excels on easy cases but breaks on one family. Scores within 0.01 of each other are tied. Ties are broken by mean hidden-suite runtime, then by a reproducibility review.

10.6 Submission format

Upload one ZIP file to the Google Classroom assignment *SPINN Bonus Challenge (optional)* by Monday 2 November 2026, 23:59 IST. The student kit attached to that assignment holds the evaluator, the starter file and the full specification. The ZIP contains:

1. `submission.py`, defining `solve_problem(case)`, which returns an $N \times N$ array with `u[j, i]` $\approx u(x_i, y_j)$. The `case` dictionary holds `x`, `y`, `f`, `boundary_values`, `run_seed` and two callables, `f_fn(X, Y)` and `g_fn(X, Y)`, which evaluate the forcing and the boundary data at any points. A PINN can use them to sample collocation points anywhere; a numerical solver can use them to work on a finer grid than the one it must answer on. Start from the supplied `starter_submission.py`.
2. `requirements.txt`, pinning every package version.
3. `README.md`: the method, the exact reproduction command, the hardware used and the expected runtime.
4. A short results note (2–3 pages): the public metrics with five-run statistics for stochastic methods, one convergence or stability check, runtime, limitations, and citations.
5. Any trained weights, together with the code and data-generation procedure that produced them.

10.7 Hardware and runtime°

The official evaluation runs on CPU, without network access, in a clean environment built from your `requirements.txt`. The time limit per hidden case is 20 minutes of wall-clock time on CPU. A method that trains for each new forcing function must fit inside that budget. For scale, a small three-layer PINN trained for 2,000 steps took 9–14 seconds per case on a four-thread laptop CPU. You may develop and pre-train on any hardware.

10.8 Integrity rules

- The solver must compute $\hat{u}$ from the supplied inputs (`x`, `y`, `f`, `boundary_values`, `f_fn`, `g_fn`). Returning stored answers, branching on a case's identity, or looking up known solutions disqualifies the submission.
- Do not access the evaluator's private files, the network or the hidden seed. Do not assume $N = 129$.
- Seed every random-number generator from `run_seed`. The teaching team may re-run any submission and inspect its code.
- Cite the papers, code and ideas you build on, including the proposal itself.

10.9 Eligibility and award

The challenge is open to third-year, fourth-year and PhD students only. The award is fixed: a submission that passes Stages 1 and 2 adds 3 marks to its author's final IML course total for a third-year student, 2 marks for a fourth-year student and 1 mark for a PhD student. "Passing" means meeting all four Stage 1 targets on the public reference case and the Stage 2 eligibility limits on every hidden case, under the integrity rules above. Every passing submission earns the award; there is no cap on the number of awards. The Stage 3 ranking is published for recognition and carries no further marks. Nothing in this challenge exempts a student from any part of the core project's assessment.

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